Working Paper – Ana Patrícia Veloso

'Peripheral-ish' regions: Moving beyond static concepts in green technology diversification

Abstract (Should contain an unstructured abstract of 100 words)

Keywords (Should contain between 4 and 6 keywords)

Main text introduction

1. INTRODUCTION

Climate change caused by global warming represents one of the greatest challenges of the century. This phenomenon is the result of decades of human activity associated with the Industrial Revolution, population growth, energy consumption patterns and changes in land use and economic development. To control the looming environmental crisis, the European Union (EU) is committed to achieve an economy carbon neutral until 2050 (European Commission and Directorate General for Research and Innovation, 2021). The new growth strategy in Europe "aims to transform the EU into a fair and prosperous society with [...] a competitive economy" (European Commission, 2020, p.2). Since research and innovation are key for economic development in European regions (Iammarino et al. 2019), ensuring that no region is left behind is essential (Peñalosa & Castaldi, 2024).

A change in the production systems and technological capabilities must be done (Peñalosa & Castaldi, 2024). The combination of these diverse capabilities is critical to develop technologies that are needed to overcome the Green Transition (Bachtrögler-Unger et al., 2023). For this process, green diversification is one of the key factors, however, new technologies or industries do not develop from scratch, since regions diversification opportunity spaces and their trajectories differ (Pinheiro et al., 2025). Exists an unequal distribution of innovation and technological capabilities over Europe's regions (European Commission, 2022a), which determines the ability to develop technologies that are crucial for the transition (Bachtrögler-Unger et al., 2023). Therefore, the ability to foster green technological development is not uniform, differing across regions (Montresor and Quatraro 2019; Li et al. 2021; Santoalha and Boschma 2021; Balland et al., 2022).

The S3 (Smart Specialisation Strategy) provides tools that enable the replacement of obsolete specialisations (Foray et al., 2021), facilitating convergence between traditional sectors and new decarbonisation solutions (Boschma et al., 2026). Regions should invest in innovation opportunities anchored to pre-existing skills where they already have unique competitive advantage (Boschma et al., 2026), identifying context-based opportunities to develop new green technologies (Montresor & Quatraro, 2020). New economic or technological activities emerge more easily in places with productive, scientific, or technological capacities closely related to pre-existing ones (Balland & Boschma, 2021; Boschma & Frenken, 2009; Santoalha & Boschma, 2021). To select the new fields of specialisation in regions based on local capabilities the strategy uses the relatedness concept as a key principle (Iacobucci & Guzzini, 2016).

Nonetheless, one of the biggest challenges in the green development of regions remains from their difficulty in diversifying, due to the processes of dependency to which

current systems are bound (Pisano et al., 2014). Regions end up suffering from lock-in in existing specializations; embeddedness in local practices and institutions; and low knowledge production, which compromises their ability to find more competitive and sustainable paths to growth (Boschma & Martin, 2007; Boschma & Martin, 2010). This phenomenon is particularly common in peripheral regions, as they are unable to diversify due to their dependence on current systems and economic stagnation (Balland & Boschma, 2021).

For this motive, base the diversification just in local capabilities isn't sufficient, as "Other regions may also affect the development of new technologies in a given region" (Bachtrögler-Unger et al., 2023, p. 5). The linkages between regions that have lack of relevant capabilities to develop green technologies, with other regions with whom they have complementarities, would help overcome lock-ins and increases the probability of regions diversifying into new activities (Bachtrögler-Unger et al., 2023; Balland and Boschma, 2021; Grabher 1993). S3 also defends strengthening inter-regional cooperation between regions across member states, since enhances each region's innovation potential and contributes to achieving the sustainable goals in Europe (Boschma et al., 2026; Rodríguez-Pose, 2025; Bachtrögler-Unger et al., 2023). This is particularly important for less developed regions (Boschma et al., 2026; Balland and Boschma, 2021) which do not usually could connect with developed regions, remaining multiple untapped potential in inter-regional linkages in Europe (Bachtrögler-Unger et al., 2023).

With this untapped potential and shift toward green transitions, other opportunities are emerging for peripheral regions – a window opportunity to leapfrog into the forefront of environmental innovation (Penãlosa & Castaldi, 2024). Even if their contributions are more limited, all help is needed during the transition process. It has been acknowledged more broadly that being peripheral plays a role in the emergence of innovation (Pugh & Dubois, 2021; Glückler, 2014). Bachtrögler-Unger et al. (2023) point out that there are some "hidden champions", meaning that some less developed regions have capabilities in vital technologies to the transition, particularly in green field.

Despite the (unexploited) potential of peripheral regions, there is still a challenge that needs to be overcome before further studies can be carried out: the definition of periphery is imprecise and ambiguous (Glückler et al., 2023) usually with negative connotations. Just as innovation policies must avoid 'one-size-fits-all' actions (Boschma et al., 2026; Tödtling & Trippl 2005), so too must the notion of the periphery be deconstructed, rejecting general, uniform concepts that ignores the specific characteristics of each context. Glückler et al. (2023) have showed that these regions can play a strategic role within social, institutional, cultural and economic networks and that being geographically peripheric does not necessarily imply a marginal position. Far from being passive spaces, these regions can act as strategic arenas for experimentation and innovation when embedded in interregional networks of knowledge and complementarity (Glückler et al., 2023; Boschma, 2025; Vale et al., 2024).

Regardless of the urgency of the green transition, the current literature shows critical gaps that limit the effectiveness of cohesion policies. Firstly, there remains a scarcity of studies on how Smart Specialisation Strategies (S3) can effectively drive green transformation in regions with obsolete industrial bases (Montresor & Quatraro, 2020), with the vital role of inter-regional linkages in this process often being overlooked (Boschma et al., 2026). Furthermore, the geography of innovation has been dominated by a "patent bias"

– an approach that Shearmur (2012) describes as a detrimental focus on formal metrics that stigmatises peripheries as chronically lagging, while ignoring their dynamic position within knowledge networks (Pugh et al., 2024).

This paper bridges these gaps by introducing the concept of the "meta-periphery" to challenge traditional, static definitions of geographical and economic marginality. By focusing on the green technology innovation network, we demonstrate that certain regions exhibit "peripheral-ish" characteristics – meaning they can or not be peripheral to the green transition network regardless of their general economic status. Assuming that "actors as well as places can occupy core and peripheral positions" (Glückler et al., 2023, p. 8), we propose a framework that treats regions as nodes in the analysis. To operationalize this, we employ Complex Spatial Network Analysis, which allows us to model the territory not as an isolated container, but as a multidimensional and relational position within a global system. This methodological choice is deeply rooted in Grabher's (2018) argument that centres and peripheries are functionally interdependent and relationally constituted. By doing so, we shift the focus from individual entities to the relational position of territories, within the green technological field, allowing for an analysis of the systemic capacity of a territory.

To mitigate the limitations of traditional metrics, our framework incorporates multi-source relational data for Europe NUTS II, specifically we utilize patent co-inventorship, inter-regional collaboration networks and green trademarks derived from OECD REGPAT, Horizon Dashboard (Cluster 5 of Horizon Europe) and European Union Intellectual Property Office (EUIPO) databases. This multidimensional approach captures how diverse layers of knowledge interact within a territory (Peñalosa & Castaldi, 2024; Castaldi et al., 2025), acknowledging that peripherality is not an inherent quality, but a dynamic condition determined by a region's position within specific innovation networks (Glückler et al., 2023).

Consequently, our research question is: *How does the structural position of European regions within green innovation networks redefine our understanding of peripherality, and which regions emerge as the 'meta-periphery' in the context of the Green Transition?* Beyond the conceptual proposal, this paper aims to develop a context-sensitive methodological framework that enables a "fair identification" of peripheral regions. We argue that peripherality should not be a static label based on distance to the core, but a dynamic condition determined by a region's position within specific technological and innovation networks (Glückler et al., 2023; Rodríguez-Pose, 2025).

While this study specifically focuses on diversification into green technologies, we anticipate that this proposal will inspire applications in diverse contexts and stimulate new debates on peripherality. Ultimately, this critical review of European peripheries serves as a timely contribution to the ongoing dialogue between the EU's traditional, administrative definitions and the complex, relational dynamics of contemporary innovation networks.

2. PERIPHERAL REGION: A DIVERSE CONCEPT

2.1 What is a peripheral region?

The term 'peripheral region' has been highly discussed in the last years, because despite non-consensual definitions, may play an important role in some of the greatest challenges of the century. Many regions are classified as peripheral based on various criteria, but the question "what is a peripheral region?" remains unanswered (Pugh & Ding, 2026). This isn't as easy to answer as it seems, since there are various ways of classifying

these regions but no consensus. In addition to the fact that this concept remains "fuzzy" (Pugh & Dubois, 2021), also exist a "bad talking" associate with peripheral regions, which perpetuate the negative discourse (Glückler, 2023) and a paternalistic view. The terminological imprecision is an obstacle to the implementation of policies that aiming combating contemporary challenges, like green transition (Pugh & Ding, 2026; Peñalosa & Castaldi, 2024; Rodríguez-Pose & Ketterer, 2020). Another challenge in this process it's the idea that all peripheries are the same and that simply defining them geographically is enough. This isn't true since it's they're also a social and emotional construct (Pugh & Dubois, 2021; Glückler, 2023; Sedmak,2026).

Drawing on the Oxford Dictionary, Glückler et al. (2023) notes that the term periphery is defined as "the outer edge of a particular area", based just in distance and accessibility, as it has been used mostly in geography (Malecki, 1991). Others may describe this region associated with systemic disadvantages, being far from neutral definition – *eg.* "refer to regions that fall outside the labour markets of large cities or metropolitan regions" (Nilsen et al., 2023). The periphery comes associated to theories of economic polarization and political marginalization (Sedmak, 2026; Pugh and Ding, 2026) or a "structural condition of lower economic density and industrial diversification bottlenecks" (Vale et al., 2024, p.118). Being most of the time imply as the "non-core" that is shaped by social exclusion and the feeling of being left behind (Martin, 2024). Grabher (2018), says that this term aggregate in a residual category, deficient places with lack of typical qualities that promote innovation, no matter their historical and geographical specificities.

Furthermore, in the context of regional inequalities, peripherality is increasingly defined by the inability to adjust to new economic modes, such as digitalization and the green transition (De Ruyter et al., 2021; Rodríguez-Pose & Ketterer, 2020), suggesting that these regions are at the forefront of modern socio-political disruptions. To overcome this reductive view of the centre-periphery dichotomy, it is necessary to break down the various layers that constitute these territories (Pugh & Ding, 2026).

We begin this work, by decomposing the term "peripheral regions". Pugh and Ding (2026, p.234) specify a region, from the perspective of regional economic geography, as "the sub-national space". Assuming that this term should not be understood as a static entity and it should depend on the analytical lens adopted.

Given this perspective, while focusing on economic geography, we acknowledge the multidimensional and fluidity traits of regions. These traits are fundamental when transposed to the study of peripheries, which depends on the domain we are talking about. Peripheries shouldn't be assumed as a fixed or static trait, but a porous and fluid position "shifting across the political, geographical, and institutional contexts." (Öberg, 2026, p. 438).

For most of the studies a periphery is a position of disadvantage or isolation in relation to a centre of power (Glückler et al., 2023; Shearmur, 2017). The literature incorporates various forms of disadvantage: from non-innovative and dependent remote locations to economically and politically marginalized areas. However, to fully understand the potential of these regions, it is necessary to move beyond this normative nature of the peripheral condition.

Glückler et al. (2023) propose a significant shift by stripping peripherality of its normative and negative connotations. Due to the fluid nature of the concept, they argue that

it is important to distinguish between locational and network positionality, assessing the peripherality condition as relational instead of inherent trait. This perspective demonstrates that a peripheral region is not a place defined by what it lacks, but rather a relational position of distance, dispersion, and disconnection within a specific field. For this reason, a periphery is a position characterized by fewer realized interconnections between places, being only sparsely connected to the network core (least network density) and far from the geographic core (Glückler et al., 2023). By redefining the concept as a neutral position in a field, the authors open a space of opportunity. This relational perspective is synthesized in a matrix (2x2) that crosses spatial context (agglomerated vs. dispersed) with network position (core vs. periphery). This framework allows for a more nuanced classification: a region may be geographically dispersed (locational periphery) but simultaneously hold a "core" position within a specific knowledge network due to its strategic linkages. By applying this matrix, peripherality stops being a static label of failure and becomes a dynamic structural condition.

This structural dynamism is further supported by Sedmak (2026), who argues that periphery should not be seen as a static state, but as a continuous process of peripheralization. This process underpins the relational isolation theorized by Glückler et al. (2023) and materializes through four dimensions: the physical fragmentation of the territory (spatial), the economic dependence on external models (structural), the absence of a political voice (socio-political), and the devaluation of local knowledge and identity (epistemic) (Sedmak, 2026). This framework already makes a significant contribution as it moves beyond an oversimplified view of peripheries toward an understanding of peripheralization as a multidimensional constructed process.

This multidimensional nature is further reinforced by Peñalosa and Castaldi (2024), who argue that peripherality in the context of green transition "encompasses multiple dimensions that are interconnected" – geographic, socio-economic, and innovation – that together determine a region's knowledge absorptive capacity. Their work suggests that the periphery is a dynamic frontier where structural bottlenecks and network connectivity collide.

With this in mind, we define peripheral regions as sub-national spaces that within a given context, exhibit the condition off being distant, dispersed and disconnected across the multidimensional factors essential to the phenomenon. We designate it as the meta-periphery. These are "peripheral-ish" regions are territories that are seen as peripheral in some contexts, but in specific others are not.

Despite conceptual definitions it becomes evident that the heterogeneous nature of peripheral regions requires a systematization of the different dimensions in which these regions have been identified. This is essential to understanding that "periphery" is not a static state, but a condition manifesting in multiple ways depending on the context. In this regard and given the vast body of literature on the subject, the next review highlights only the theoretical frameworks and empirical applications considered most relevant to this matter. This provides a clear picture of how these regions have been defined.

2.2 How have these regions been defined empirically and perceived?

The " Traditional" Peripheries

The concept of periphery, historically, has been dominated by negative connotations (Pugh & Dubois, 2021), where territories are described by their lack or absence of attributes (Glückler et al., 2023; Pugh et al., 2024) and guided by the diversity of names and terms used for them. To unpack how the concept is currently being applied in Europe, our research draws on the main empirical studies. We identify six different dimensions (Table X) used to define peripheries empirically: Geographic, Economic, Functional, Institutional, Demographic and Technological and Digital.

The traditional approach to peripherality is defined by geographic distance, focusing on territories structurally isolated from major economic cores (Espínola & Cravidão, 2025). In the European Union, this is exemplified by the Outermost Regions (ORs), where permanent constraints such as remoteness, insularity, small size, and challenging topography dictate their peripheral status (European Union, 2012).

The Inner Peripheries (ESPON, 2018) provides a more complex layer to the traditional geographic view. Here, peripherality is not about kilometres from a capital, but about functional connectivity. Territories which, although situated in the heart of Europe or close to prosperous centres, can be peripheralized through macro-remotism (reduced accessibility to large agglomerations and dominant economic centres) (Pugh & Dubois, 2021). These regions are defined not by their location, but by the disproportionate effort required to operate from them, leading to a state of functional exclusion (Sedmak, 2026).

Traditionally, the EU's Cohesion Policy (European Union, 2021) has quantified this through an Economic Periphery lens, using the 75% GDP threshold to define less developed regions. Yet this static metric is increasingly seen as insufficient. As Rodríguez-Pose and Wilkie (2018) warn, the core issue for these lagging regions is not just low income, but a structural inability to generate innovation.

The inability to generate innovation, positioning regions as subaltern in the global value chain. This subalternity is often mirrored by Institutional Peripheries. Institutional marginalisation it can be expressed with the European Quality of Government Index and characterised by an institutional vacuum, in other words low state capacity, inefficiency or corruption (Charron et al., 2014). Sedmak (2026) links this to a lack of political influence. Regions with weak subnational institutions (Charron & Lapuente, 2018) have less say in European decision-making. In this sense, the periphery is a political position where the lack of agency prevents the strategic absorption of funds, perpetuating a cycle of marginalization (Sedmak, 2026).

The concept of Shrinking Regions (ESPON, 2018) represents a Demographic Periphery, characterised by extreme ageing and a brain drain. These regions offer limited opportunities, which generates the migration of skilled young people (Sedmak, 2026). This permanent population decline (Martinez-Fernandez et al., 2012) is the critical sign of peripheralization, where the lack of a critical mass leads to further disconnection to central flows, reinforcing structural disadvantages.

This disconnection is further amplified by the emergence of a Technological and Digital Periphery, where gaps in infrastructure and skills create new layers of exclusion

(European Commission, 2022b). These regions suffer from deficiencies in human capital, digital integration, and public services, placing them in a position of structural deficit (Moraliyska, 2024). Crucially, this condition affects regions that, despite being geographically close to economic centres, remain functionally isolated due to a low capacity for creating digital content and a systemic dependence on transnational platforms (Szczepanik et al., 2020). Consequently, the lack of high-speed connectivity and digital literacy places these regions in a position of mere consumers rather than producers in the modern economy, reinforcing a narrative of peripherality defined by technological marginalization.

These different forms of peripherality do not exist in isolation but rather create a cycle in which vulnerabilities feed into one another, reinforcing the isolation of these regions and the associated normative connotations.

Periphery Dimension	Criteria	Central Concept	Perspective	Reference
Geographic	Remoteness, insularity, and permanent natural constraints (topography/climate).	Outermost Regions (ORs)	Peripherality as a structural problem of accessibility and distance from the core.	European Union (2012)
Economic	Economic convergence based on productivity and income (GDP < 75% of EU average).	Less Developed Regions	The periphery as a problem of lack of economic convergence.	European Union (2021)
Functional	Lack of access to Services of General Interest (health, education, transport) within prosperous spaces.	Inner Peripheries	The periphery as a problem of lack of functional connection.	ESPON (2018)
Institutional	Weak state capacity, high corruption, or inefficient bureaucracy (European Quality of Government Index)	Institutional Void	The periphery as a problem of lack of governance quality.	Charron et al. (2024)
Demographic	Social viability of the territory; extreme aging and "brain drain" (loss of young talent).	Shrinking Regions	The periphery as a problem of lack of human capital and social vitality.	ESPON (2017)
Technological & Digital	Absence of high-speed broadband and digital literacy. Knowledge is not produced, only consumed.	Digital Divide	The periphery as a problem of functional exclusion and technological dependence.	European Commission (2022b)

The Periphery as an opportunity: green technology innovation

Traditional studies, continue to view the periphery as a place defined by the lack of something. This is particularly evident in the Geography of Innovation, which has historically marginalized these regions by characterizing them as laggard, non-innovative (Glückler et al., 2023; Shearmur, 2017; Tödtling & Trippl, 2005). However, this normative conceptualization is no longer sufficient. Peripheral regions are not necessarily

disadvantaged in all contexts; instead, they can play a vital role in social, cultural, and economic networks (Glückler et al., 2023).

Despite significant structural challenges, such as dependence on polluting industries, institutional lock-in and a lack of endogenous knowledge (Tödtling & Trippl, 2018; Grillitsch & Hansen, 2019; Vale et al., 2024), recent literature suggests that the periphery holds latent potential for innovation in green technologies. A more fluid and relational perspective is thus emerging, viewing the periphery as a space of opportunity for a sustainable future (Pugh & Dubois, 2024). This opportunity lies largely in the ability of regional actors to integrate pre-existing capabilities into global innovation systems (Binz & Anadon, 2018). In this way, peripheral regions are no longer seen as mere passive beneficiaries of technology, but rather as spaces where technological innovations and new green pathways can emerge (Yu & Gibbs, 2018; Vale et al., 2024).

Peripherality should not be seen as a sealed fate determined by geography, but rather as a deliberate choice to shield creativity and dissenting ideas from the mimetic pressures of the urban mainstream (Grabher's, 2018). Peripheral regions are identified by Grabher (2018) as "outsiders", by being protected from the rigid conformity of the core, possess the freedom to develop the radical innovations required for the green transition. The same happens in Bachtrögler-Unger et al. (2023) study, where they identify some "hidden champions", that corresponds to less developed regions that hold valuable capabilities in technologies transition. For this, the periphery should not be understood as a fixed or normative attribute (Glückler et al.,2023).

In the current context of globalization, the European green innovation network is best understood through connectivity capacity and the strength of knowledge assets rather than mere physical proximity. As Hoekman (2012) demonstrates, the geography of collaboration allows knowledge production to transcend regional boundaries, offering international co-authorship and research networks as efficient mechanisms to overcome geographical distance. Consequently, peripheral regions are redefined not by their location, but by their failure to anchor themselves in these relational flows. They are characterized by low centrality and a lack of institutional and organizational proximity, which hinders their integration into global knowledge systems and keeps them on the margins of the green transition (Glückler et al., 2023; Hoekman, 2012).

This perspective, show us the opportunity that can be allowing peripheral regions to overcoming the relevant capabilities that are often missing, through complementary external collaborations (Bachtrögler-Unger et al., 2023; Balland & Boschma, 2021). However, the nature of this diversification is inherently shaped by the region's development stage, while advanced regions have the capacity to take risks and leap into highly complex, unrelated technologies; less developed regions face a more path-dependent reality (Hellinga et al., 2025). In these territories, diversification is closely tied to the existing technological profile. Consequently, inter-regional linkages become vital for acquiring the complementary skills needed to cross this path, particularly in low-complexity green technologies where peripheral regions hold significant untapped potential (Pinheiro et al., 2025; Bachtrögler-Unger et al., 2023). This process reveals a unique relational strength: beyond the high propensity for cross-border cooperation – where less developed regions lead in share of linkages – these territories can strategically leverage this connectivity to focus on research and innovation specifically targeted at the green transition (Bachtrögler-Unger et al., 2023).

The periphery is now based on three fundamental perspectives: geography, innovation capacity and socio-economic context (Peñalosa and Castaldi, 2024). The innovation component incorporates access to key flows and the relational centrality of the European green innovation network – which is defined by integration into collaborative networks that enable the capture of innovation capacity (Peñalosa and Castaldi, 2024). However, the reliance on traditional metrics like patents often obscures the real picture. Patents are subject to a significant "urban bias," over-representing large city-regions due to a higher patent-filing propensity driven by corporate headquarters and administrative density (Taalbi & Martynovich, 2024; Shearmur, 2012, 2017). Relying solely on patent data as a proxy for innovation is problematic, particularly for peripheral regions, instead it should be complemented with other sources when analysing non-urban patterns (Taalbi & Martynovich, 2024).

In this context, complementary metrics such as participation in international research consortia (e.g. Horizon Europe) and co-inventor links become essential to full address the spectrum of innovation in peripheries. These relational indicators reveal "invisible" innovation pattern that patents often fail to record (Castaldi et al., 2025), providing a more accurate map of the systemic capacity within the European green innovation network of these regions.

The European green innovation network is structured by connectivity capacity and the strength of knowledge assets, with peripheral regions being those which, regardless of their location, remain on the margins of these relational and institutional flows.

Ultimately, the European green innovation network is structured by connectivity and knowledge assets. Peripheral regions are those that, regardless of location, remain on the margins of these flows. To tackle the widening gap in Europe, it is crucial to identify the opportunities in low-income regions and remove the barriers blocking them from progressing (Bachtrögler-Unger et al., 2023). Consequently, the following chapter details the methodology designed to map these relational dynamics of the *"meta-periphery"*.

Materials and methods

3. DATA AND METHODS

4.RESULTS, DISCUSSION

5.CONCLUSION INCLUDING POLICY RECOMMENDATIONS

DECLARATION OF INTEREST STATEMENT

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Table(s) with caption(s) (on individual pages)

Figures

Figure captions (as a list)